

Using GAN Classifiers to Detect Deepfakes

Jeeven Balasubramaniam

Rice University

6100 Main St, Houston, TX 77005

jb310@rice.edu

Prathamesh Swar

Rice University

6100 Main St, Houston, TX 77005

ps165@rice.edu

Mohamad Kreidieh

Rice University

6100 Main St, Houston, TX 77005

mk232@rice.edu

Syed Fazail Haider

Rice University

6100 Main St, Houston, TX 77005

sh253@rice.edu

Abstract—The rapid advancement of artificial intelligence has enabled the creation of highly realistic synthetic media, commonly known as deepfakes, which pose serious risks to trust, privacy, and information integrity. To preserve the search for truth, we must develop automated systems to distinguish between authentic and forged data. This project explores the use of Generative Adversarial Networks (GANs) that implement Convolutional Neural Networks (CNNs) for detecting fake images by leveraging the discriminator’s ability to distinguish real from generated samples. We conduct preliminary experiments on the MNIST dataset (handwritten 0-9 digits) to validate the effectiveness of GAN-based detection in a controlled setting, followed by experiments on more complex datasets, including anime-style images and the CelebFaces dataset (images of celebrity faces). Our results on MNIST demonstrate that not only does the generator create highly realistic images of handwritten digits but also the discriminator has small loss values meaning it can reliably differentiate real and synthetic images, providing a foundation for extending GAN-based detection methods to more realistic and challenging image domains. Due to greater complexity of the other two datasets, the generator and discriminator losses though small are greater than the one for MNIST but show considerable possibility for effective use when provided sufficient computational resources.

Index Terms—GAN, Generative Adversarial Network, Deepfake Detection, Discriminator, Synthetic Media

I. BACKGROUND/MOTIVATION

In recent years, advances in artificial intelligence have enabled the creation of highly realistic synthetic media, commonly referred to as deepfakes. Deepfakes are images or videos in which a person’s face, voice, or other content is digitally manipulated or entirely synthesized, often using Generative Adversarial Networks (GANs) or similar generative models [1]. These manipulations can be so convincing that even humans struggle to distinguish real content from fake, posing serious risks to society. Deepfakes have been exploited for spreading misinformation, identity theft, political manipulation, or malicious pranks, making reliable detection methods an urgent and important challenge [2], [3]. Understanding the subtle artifacts of generated media is therefore critical for maintaining trust in digital content.

Traditional detection approaches rely on supervised classifiers, such as convolutional neural networks (CNNs) or transformers, trained on datasets like FaceForensics++ [2] or

the Deepfake Detection Challenge (DFDC) dataset [3]. While effective on benchmark datasets, these models often struggle to generalize to unseen deepfakes or to media that has been compressed, resized or altered post-generation, highlighting the need for more robust approaches that can capture subtle inconsistencies beyond standard learned features [4].

An alternative and promising approach is to leverage GANs not for generation, but for detection. In a GAN, the discriminator learns to distinguish real from generated content by identifying subtle differences in image statistics and textures [1]. GAN-based anomaly detection methods, such as f-AnoGAN [5], use reconstruction errors from generators trained on real images to identify unnatural patterns in fake content. This approach transforms a tool traditionally used for content creation into a forensic instrument for content verification.

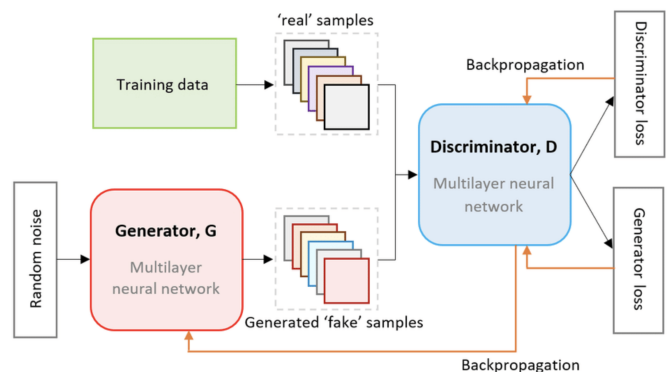


Fig. 1. GAN Architecture

The motivation for this project is twofold. First, it explores a novel application of GANs for detecting fake media across datasets of increasing complexity. We begin with MNIST as a preliminary experiment to validate the approach, and then progress to anime-style images and the CelebFaces dataset to evaluate detection performance on more realistic and challenging data. Second, it addresses the societal importance of identifying manipulated media, contributing to research on more robust, interpretable, and generalizable detection

methods. By investigating GAN-based detection, this project aims to provide a foundation for resilient detection systems and a deeper understanding of the artifacts that distinguish real from synthetic content.

II. EXPERIMENTATION AND MODEL ARCHITECTURE

A. Data Preprocessing and Augmentation

To ensure the stability of the GAN across domains, distinct preprocessing pipelines were established for each of the three datasets (MNIST, CelebFaces, and Anime Faces).

TABLE I
DATASET COMPARISON

Feature	MNIST	Anime	CelebA
Content	Handwritten Digits (0-9)	Anime Faces	Celebrity Faces
Images	70,000	21,551	202,599
Size	28×28	64×64	178×218
Channels	1 (Gray)	3 (RGB)	3 (RGB)
Classes	10 Digits	N/A	40 Attributes

1) *Normalization*: A critical step for Deep Convolutional GAN (DCGAN) stability is matching the input data distribution to the Generator's output activation. Since our Generator utilizes a $\tanh$ activation function in its final layer (outputting values in the range $[-1, 1]$), all training images were normalized from the pixel range $[0, 255]$ to $[-1, 1]$ using the transformation:

$$x_{norm} = \frac{x - 127.5}{127.5} \quad (1)$$

2) Resizing and Formatting:

- **MNIST**: As the images are single-channel ($28 \times 28 \times 1$), we reshaped the flat vectors into 3D tensors. Unlike the face datasets, MNIST required a modified architecture (removing one upsampling layer) to accommodate the lower resolution.
- **Anime Faces**: The dataset images were pre-processed to $64 \times 64 \times 3$. No additional resizing was required, allowing for direct integration into the CelebA-optimized architecture.
- **CelebFaces**: The original images (178×218) contain significant background noise. We applied a center crop of 108×108 to isolate facial features, followed by a resize to $128 \times 128 \times 3$.

B. Model Architecture (DCGAN)

We adopted a DCGAN architecture, replacing standard fully connected layers with Convolutional layers to better capture spatial hierarchies as we know CNNs capture the feature maps in images with great accuracy.

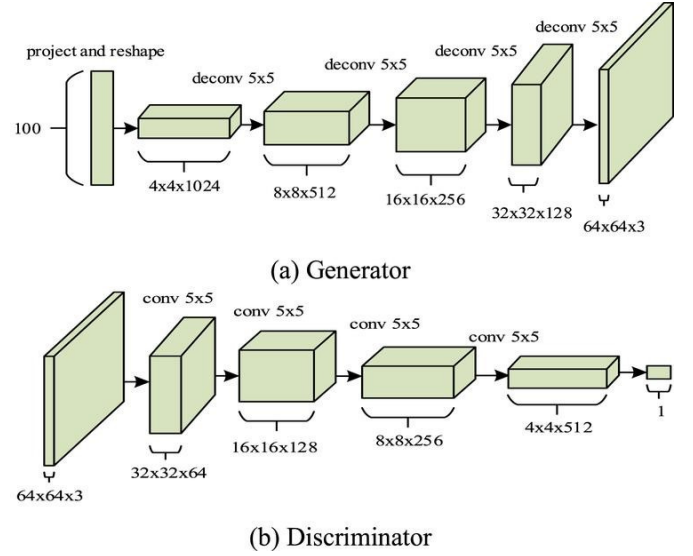


Fig. 2. Typical DCGAN Architecture

1) *The Generator (G)*: The network accepts a latent vector z ($dim = 100$) sampled from a normal distribution. This is projected via a Dense layer and reshaped into a $4 \times 4 \times 256$ feature map (or 7×7 for MNIST). The signal passes through a series of Conv2DTranspose (deconvolution) layers with strided upsampling. Each layer utilizes Batch Normalization to stabilize learning and LeakyReLU activation, culminating in a $\tanh$ activation to generate the final image.

2) *The Discriminator (D)*: This acts as a binary classifier taking an image as input. It uses strided Conv2D layers to downsample the image resolution while increasing feature depth. To prevent the Discriminator from overpowering the Generator early in training, we utilized LeakyReLU ($\alpha = 0.2$) and Dropout (0.3).

C. Criteria for Results

- 1) Stable Generator and Discriminator loss curves over training
- 2) No mode collapse: diversity across generated digits for MNIST, varied facial/hair features (color, size, expression of face) and orientations for generated Anime and Celebrity Faces images
- 3) Qualitative visual inspection by humans leads to maximum possible realistic generated images for all datasets
- 4) Optional: compute a simple FID score when transitioning from less complex MNIST to more complex Anime and CelebFaces datasets

D. Experimental Iterations and Challenges

Our experimentation process was iterative, moving from simple baselines to the complex DCGAN required for high-fidelity face generation. We encountered several common GAN failure modes and addressed them through architectural and hyperparameter adjustments.

1) *Latent Dimension Tuning*: We initially experimented with a high latent dimension ($dim = 100$) to capture complex facial features. However, we observed that this led to slower convergence without a noticeable improvement in generated image quality. The high dimensionality increased the number of parameters in the Generator’s initial dense layer significantly. We reduced the latent dimension to 32, which proved sufficient for encoding facial features while significantly reducing the computational load and training time.

2) *Learning Rate Sensitivity*: The Adam optimizer learning rate was a critical hyperparameter. An initial rate of 10^{-3} resulted in very high loss values for both the Generator and Discriminator, leading to unstable oscillations where neither network could learn effectively. By reducing the learning rate to 10^{-4} (0.0001), we achieved much lower and more stable loss values, allowing the model to converge on realistic features. With $LR = 10^{-3}$ for CelebFaces, the loss values for 25 epochs were far from ideal (G Loss was 93.7 even after 25 epochs of training) as shown below:

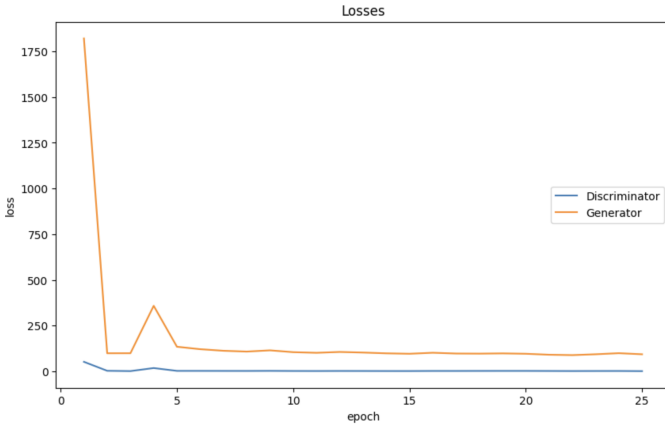


Fig. 3. CelebFaces Loss with *Adam* $LR = 10^{-3}$

3) *Discriminator Dominance and Regularization*: Early in training, we observed that the Discriminator would overpower the Generator, with its loss dropping to near zero almost immediately. When the Discriminator is too strong, the gradients passed to the Generator vanish, halting learning. To counter this, we introduced two key regularization techniques:

- 1) **Label Smoothing**: We replaced hard labels (1.0) with soft labels (e.g., 0.9) for real images.
- 2) **Dropout**: We added Dropout layers ($rate = 0.3$) to the Discriminator.

These changes successfully limited the discriminator, allowing the generator sufficient time to learn structural features.

4) *Training Duration and Resource Constraints*: We found that the optimal training duration varied significantly by dataset. For the MNIST dataset, we trained for 300 epochs while for Anime dataset, we trained for 50 epochs to fully resolve the sharp lines characteristic of the style. However, for the CelebFaces dataset, we observed negligible improvement in visual quality beyond 20 epochs. Given significant resource

constraints, where a single epoch on CelebFaces required approximately 15 minutes of GPU time, we capped training at 20 epochs. This provided a balance between image fidelity and computational feasibility.

5) *Unsuccessful Experiment*: Initially, we attempted to generate celebrity faces using a Multi-Layer Perceptron (Dense layers only). This approach was largely unsuccessful because the model suffered from **Mode Collapse**, where the Generator produced a single, blurry “average” face regardless of the input noise. Dense layers failed to capture local spatial correlations (e.g., the relative position of eyes and mouth), and the parameter explosion required to map a 100-dim vector to 12,288 pixels ($64 \times 64 \times 3$) led to poor optimization landscapes.

E. Final Hyperparameter Configuration

The most successful configuration, used to generate the final results presented in this report, is as follows:

- **Framework**: TensorFlow/Keras
- **Optimizer**: Adam ($\alpha = 0.0001, \beta_1 = 0.5$)
- **Loss Function**: Binary Cross-Entropy (with logits)
- **Batch Size**: 256 (MNIST), 128 (Anime/CelebFaces)
- **Latent Dimension**: 32
- **Epochs**: 300 (MNIST), 50 (Anime), 20 (CelebFaces)

III. RESULTS

We evaluate the model based on training stability (Loss Convergence) and the perceptual quality of the generated images (Visual Assessment). Here are the losses for each of the 3 datasets:

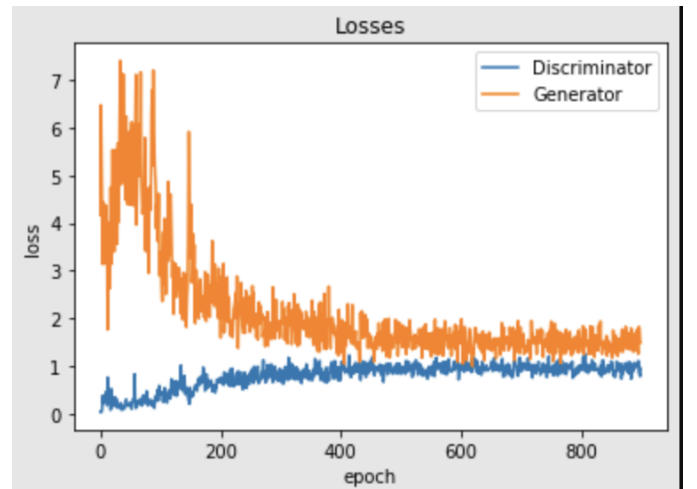


Fig. 4. Graph for MNIST Losses

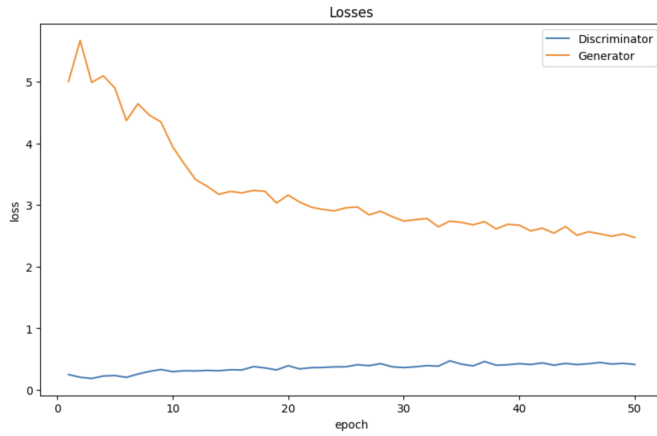


Fig. 5. Graph for Anime Faces Losses

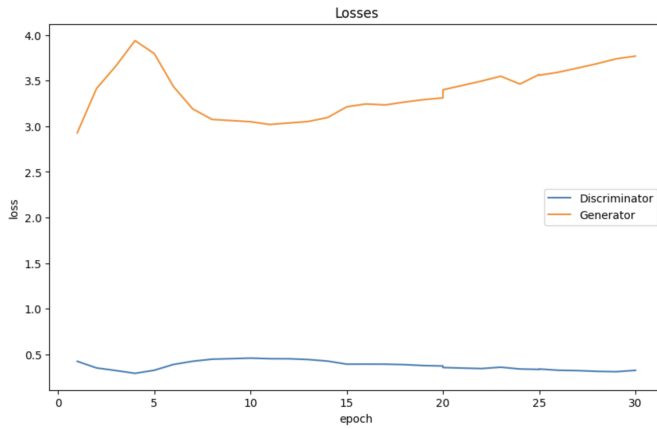


Fig. 6. Graph for CelebFaces Losses

TABLE II
COMPARATIVE ANALYSIS OF DCGAN PERFORMANCE

Dataset	Training Stability	Discriminator Status	Generator Trend
MNIST	Volatile (Noisy)	Challenged (Rising)	Oscillating
Anime	Smooth Convergence	Balanced	Steadily Improving
CelebFaces	Very Stable	Dominant ($Loss < 0.5$)	Slowly Rising

The comparative analysis demonstrates that the DCGAN architecture's performance is highly dependent on dataset complexity and domain. The key findings are summarized below:

- **Complexity vs. Convergence:** Simpler, lower-dimensional datasets like MNIST and Anime exhibited the most desirable training behaviors. These were characterized by either smooth convergence or "healthy"

adversarial volatility, where neither network overpowered the other, maintaining a balanced minimax game.

- **Photorealism Challenges:** In contrast, the photorealistic CelebFaces dataset presented a significant challenge. The training logs indicated discriminator dominance (with $D_{loss} < 0.5$), suggesting that the model found it easier to distinguish generated artifacts in complex facial data than to generate convincing photorealistic features.

Below are the improvements in the generated images for each dataset:

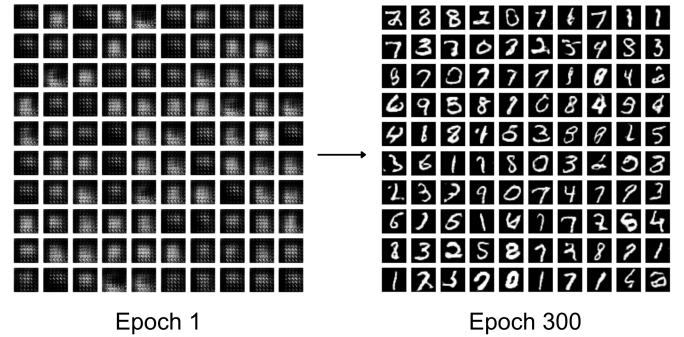


Fig. 7. Improvement in Generated MNIST Image

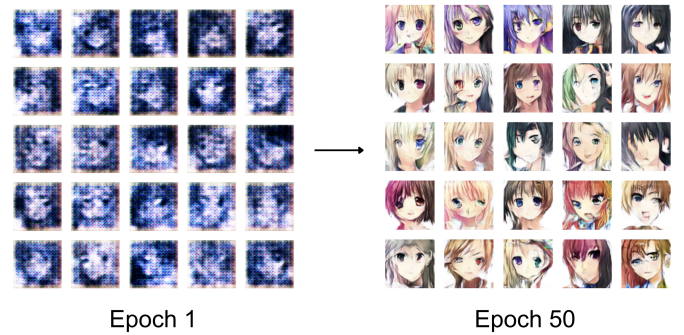


Fig. 8. Improvement in Generated Anime Image

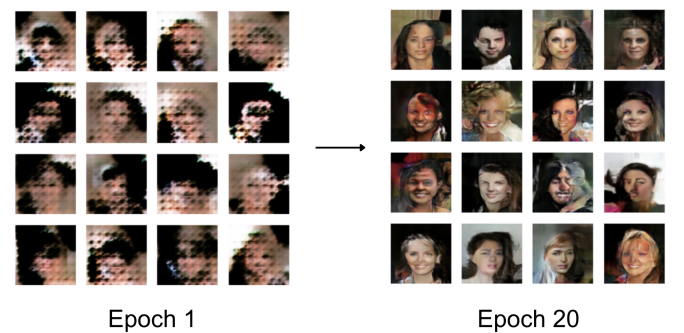


Fig. 9. Improvement in Generated CelebFaces Image

Summary of Generated Image Results:

- **Visual Convergence & Stability:** In all three scenarios, the model successfully transitions from random noise (Epoch 1) to coherent, high-fidelity images. This confirms the stability of the GAN architecture used.
- **Domain Versatility:**
 - *MNIST*: The model mastered simple geometric structures (handwritten digits) by **Epoch 300**, producing clear, unambiguous numbers.
 - *Anime Faces*: The model handled stylized features and coloring successfully. By **Epoch 50**, it generated diverse anime characters with distinct hair colors, eye shapes, and facial structures.
 - *CelebFaces*: The model tackled photorealistic features. By **Epoch 20**, it evolved from indistinguishable blobs to recognizable human faces with varied skin tones, lighting conditions, and facial orientations.

Novelty of Results Emphasized: In the context of Deepfake detection research, the novelty highlighted by these figures includes:

- **Rapid Convergence on Complex Data:** The most notable result is the speed at which the model learned photorealistic features. While simple digits took 300 epochs to perfect, the complex human faces were recognizably formed by only **Epoch 20**. This suggests a highly efficient training architecture capable of generating deceptive photorealistic “deepfakes” very early in the training cycle.
- **Creation of Diverse “Fake” Datasets:** The results emphasize the model’s ability to generate significant diversity (different genders, styles, and angles) rather than suffering from “mode collapse” (where a GAN produces the same image repeatedly). This ensures the deepfake detector is tested against a wide variety of synthetic examples.

IV. DISCUSSION AND CONCLUSION

A. Summary of Findings vs. Expectations

The results of our implementation were largely consistent with theoretical expectations for GANs, though the sensitivity of the training dynamics was more pronounced than anticipated. Our experiments revealed that architectural depth and hyperparameter settings must be strictly tailored to the complexity of the data. While MNIST converged rapidly with a shallow network, the high-frequency details in Anime and CelebFaces required significantly deeper networks and careful regularization.

Contrary to our initial hope that a higher latent dimension ($dim = 100$) would yield better features, we found that a smaller dimension ($dim = 32$) was sufficient and computationally more efficient. This suggests that the manifold of human faces can be effectively compressed into a relatively low-dimensional space.

B. Influence of Experimentation on Results

The iterative refinement of our model was the deciding factor in our success. Our early baseline models failed due to Mode Collapse and the “vanishing gradient” problem. The introduction of Dropout (0.3) in the discriminator and the reduction of the Adam optimizer’s learning rate to 10^{-4} were crucial interventions. Early experiments without these adjustments resulted in the Discriminator “memorizing” the training set, forcing the Generator to produce garbage noise. By artificially weakening the Discriminator, we achieved a stable Nash equilibrium where both networks improved in tandem.

C. Implications

The successful generation of photorealistic faces and high-quality anime characters demonstrates the power of DCGANs for creative content generation and data augmentation. The ability to generate infinite variations of anime faces implies that GANs can be used to artificially expand small datasets for downstream tasks where labeled data is scarce. Furthermore, our findings imply that “standard” hyperparameters found in literature are rarely one-size-fits-all; specific domain constraints dictate the need for custom regularization strategies.

D. Future Directions

If this project were to be continued, the next logical step would be to pivot from generation to detection. We propose using our trained Discriminator as a Deepfake Detector. Future work would involve evaluating the Discriminator as a binary classifier on a held-out test set of real celebrity images versus deepfakes generated by other state-of-the-art models. This would test the model’s generalization capabilities. Additionally, implementing Conditional GANs (cGAN) would allow for controlled generation (e.g., specifically generating “smiling” faces), adding a layer of utility beyond random sampling.

V. CODE AVAILABILITY

This is the link to the [GitHub Project Repository](#).

VI. GROUP WORK

All of us brainstormed ideas, decided the scope and logistics of the project and used our Colab and other GPU credits for training the models.

- Jeeven worked on the GAN for CelebFaces dataset, fine-tuning to adjust the dataset, ensuring the best hyperparameters were used given the resource constraints & image outputs and losses were saved.
- Prathamesh developed the GAN for Anime dataset from the MNIST model and fine-tuned the original model to fit the dataset and get improved generated images with smaller loss values as training progressed. He also facilitated in implementing the CelebFaces dataset.
- Mohamad worked on the model for MNIST CelebFaces dataset, ensuring generated images and loss values were being saved. He also assisted with the poster and report.

- Syed worked on the MNIST dataset model to formulate the basic model, designed the poster, and drafted the project report. He also ran the CelebFaces model with the inefficient optimizer learning rate which was a huge learning for us.

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